Renormalisation of Wilson Loops in Four-Dimensional Yang–Mills Theory

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Bałaban's renormalisation group (RG) construction gives a gauge-fixed lattice measure and a corresponding RG trajectory for Yang–Mills at every finite ultraviolet cutoff. However, the gauge transformations introduced at successive steps accumulate, and seem to hinder convergence of the resulting gauge-fixed fields as the cutoff is removed.[1]

We therefore seek the continuum theory through gauge-invariant observables. The natural candidates are Wilson loops and pointwise traces of powers of the field strength, which present a second difficulty: in dimensions greater than two, they are ultraviolet singular and cannot be treated as ordinary continuum random variables.

Our programme addresses this problem for correlation functions of Wilson loops in four-dimensional Yang–Mills theory. We aim to identify the scalar contributions within the RG corresponding to the perimeter and corner renormalisations known from the physics literature, and to prove ultraviolet convergence after their extraction.

An Overview of the Ideas Involved in our Method

We work at fixed finite physical volume with a compact connected simple gauge group. We allow an arbitrary but positive fine lattice spacing $\varepsilon > 0$ and a sufficiently small coupling g_ε. We perform the first RG step starting from the bare action to a lattice with spacing $L\varepsilon$, for L sufficiently large, independent of ε.

Integrating the fine fluctuations of a Wilson loop gives the schematic expansion

$$W_{\text{fine}}(C) \approx \kappa_C W_{\text{coarse}}(C) + \Delta_C \ , \qquad \kappa_C = e^{-\delta m\, P(C)} z_c^{\,4} \ .$$

[1] Lack of convergence does not contradict Bałaban's UV stability result, nor would it disprove the existence of the physically relevant part of a gauge theory in four dimensions.

For a square contour C, $P(C)$ is its perimeter, δm the perimeter counterterm, and z_c the geometric mean of its four corner factors. Extracting κ_C separates the accumulating scalar contributions from the retained-field dependence. Renormalisation means dividing by this factor and controlling the remaining correction through further steps.

We show that δm and z_c may be computed by integrating the fine fluctuations around the trivial coarse field, the identity, and that they satisfy the following bounds. Up to logarithmic factors in the bare coupling, we have

$$|\delta m| \lesssim \frac{g_\varepsilon^2}{\varepsilon}\,, \qquad\qquad |\log z_c| \lesssim g_\varepsilon^2\,.$$

Each coarse segment contributes a small amount to the perimeter exponent, but refining the lattice increases their number: the counterterm can grow at the rate ε^{-1}. The corner logarithm has no such length factor, although corner contributions can accumulate over RG steps. This is the expected ultraviolet scaling of the perimeter and corner contributions.

Extending these estimates to successive RG steps, together with bounds on the remainder, will allow us to control their accumulation under the RG. Using Bałaban's large-field decomposition and forest expansions, we show that corrections to the identity-background calculation gain an additional factor of the coarse-field curvature in small-field regions, while large-field contributions are suppressed by their small probabilities. We also obtain analogous first-step bounds for the correlation function of two square loops with no common vertices, with additional cross terms that decay exponentially with their separation.

However, our current bound on the remainder, up to logarithmic corrections, is only

$$|\mathbb{E}[\Delta_C]| \lesssim \frac{g_\varepsilon^3}{\varepsilon}$$

due to a crude per bond estimate. This is insufficient for concluding UV convergence and we are currently working on eliminating the ε^{-1} term and obtaining the correct g_ε^3 bound.

Further, $W_{\text{coarse}}(C)$ is in fact a "dressed" Wilson loop which retains parts of the large field history in the spirit of Bałaban's large field mechanism. The available first-step comparison costs order g_ε per coarse contour bond, up to logarithmic factors. Since the number of these bonds grows like ε^{-1}, we retain the dressed insertion through the RG and postpone comparison with an ordinary loop until the final IR scale.

The Programme for Iteration and the Ultraviolet Limit

The next task is to show that the insertion produced by the first RG step can be passed through further steps, while continuing to extract the perimeter and corner factors and control the remaining corrections. Parts of the loop may cross large-field regions or depend on fields outside the region being integrated. The continuation drafts develop a local representation that retains these contributions within the closed, gauge-invariant insertion, together with bounds on their dependence on the remaining fields.

This is needed to integrate those fields at the next RG step while preserving control of the corrections. The latest refinements still require checking and incorporation into the manuscript. Thus, our current goal is to extend our estimates from the bare lattice measure to the effective measures generated along the RG flow.

We must control the inherited interactions and previously generated Wilson loop corrections, so that perimeter and corner contributions can continue to be extracted and the remaining corrections satisfy the same type of bounds at every scale. The latest drafts already sketch substantial parts of this extension, but the induction is not yet complete.

Once we have proved that our Wilson loop estimates can be iterated through arbitrarily many RG steps, we aim to compare theories with successive ultraviolet cutoffs, with their bare couplings tuned to the same renormalised coupling at a fixed reference scale. The required result is a summable bound on successive differences of the renormalised loop expectations.

Draft Results and Further milestones

The latest drafts give bounds uniform in the ultraviolet cutoff and convergence along subsequences at a fixed physical stopping scale, subject to the stated inputs for the large-field steps and the relative-weight comparison. They also derive an exact formula for the difference between successive cutoffs and a geometrically small bound on the contribution of old RG steps. What remains is to control the recent-step outputs under refinement of the fine lattice, including the reconstructed backgrounds, fluctuation integrals, coupling counterterms and remainder differences. This would complete the summable comparison needed for convergence along the full sequence.

The remaining milestones are:

- Complete the verification of the written uniform iteration argument, including the large-field renormalisation steps and the required locality estimates, and incorporate it into the manuscript.

- Verify the draft's relative-weight bound for the surviving large-field regions at the stopping scale, and its consequences for uniform bounds and subsequential limits of the renormalised loop expectations.

- Complete the recent-step stability estimate in the cutoff comparison. The draft already gives the exact difference formula and a geometrically small bound on the contribution of old RG steps. What remains is to control the recent-step outputs, reconstructed backgrounds, fluctuation integrals and coupling counterterms under refinement of the fine lattice at a fixed physical scale.

- Combine this estimate with the existing reduction to obtain a summable bound on successive differences of the full renormalised loop expectations, including the remainder differences, and hence uniqueness of the subsequential limit and convergence along the full sequence.

Given current progress, we believe that these should be finished within the next few days.

Finally, we wish to mention that, if a method exists for proving the full mass gap that would be close to the existing literature, we believe that it likely lies in finding a correlation function inequality for the correlation functions of renormalised Wilson loops.

Contrast with other Efforts

We want to contrast our approach with a similar effort recently advanced by Ethan Sussman (`https://github.com/ethanwsussman/Yang--Mills-LLM-experiment`). Similar to us he seems to have focused on constructing correlation functions of gauge invariant objects, specifically he focuses on renormalising $\operatorname{tr} F^2$, with a published work showing the existence of the correlation functions of this composite operators on non-coinciding points for three-dimensional Yang Mills. However, upon completion we expect our result to fit within less than 400 pages.

We do not yet know what approach has been taken by the recently announced result by OpenAI. However, due to the structural limitations of gauge theories requiring one to apply gauge transformations at each step of the RG flow, we believe that it is likely similar to our approach.